Model-Based Denoising of Digital Mammography Incorporating Spatial Noise Correlation

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Abstract

Purpose: This study proposes and evaluates a model-based denoising framework for digital mammography (DM) that explicitly accounts for spatially correlated noise. **Methods:** Noise-free synthetic images were generated using Virtual Clinical Trials (VCTs), and spatially correlated quantum and electronic noise was simulated to produce images at 50%, 75%, and 100% of the standard dose. The simulated images were denoised using two different model-based pipelines: one assuming spatially non-correlated noise, and another accounting for spatial noise correlation. Performance was assessed using the Mean Normalized Squared Error (MNSE) and its decomposition into bias and residual noise components. Power Spectrum (PS) analysis was used to evaluate spectral characteristics, and detection performance was assessed using the detectability index (d') obtained from a model observer. **Results:** The correlation-based denoising consistently achieved lower MNSE values across all dose levels compared to the non-correlated noise approach, with improvements primarily associated with a reduction in signal loss (bias). Spectral analysis demonstrated that the correlation-based method more closely preserved the reference power spectrum, particularly at higher spatial frequencies, whereas the non-correlated noise approach exhibited spectral attenuation consistent with over-smoothing. The d' results indicated that the spatial correlation assumption for denoising achieved better detection performance. **Conclusion:** Incorporating spatial noise correlation into model-based denoising improves quantitative and spectral performance in DM images, particularly under low-dose conditions. These findings highlight the importance of accurate noise modeling for preserving fine details, such as microcalcifications.

1 BACKGROUND AND PURPOSE

Digital mammography (DM) plays a central role in breast cancer screening programs in several countries[1]. However, the diagnostic performance of DM is directly dependent on image quality, which is strongly influenced by noise and radiation dose[2;3].

Noise in digital mammography systems mainly arises from two sources: quantum noise, associated with the stochastic interactions of X-ray photons with the detector, and electronic noise, related to the readout process of the system[4;5]. Additionally, in indirect conversion systems based on scintillators, where X-rays are converted into visible light during image acquisition, spatial correlation between pixels is introduced, modifying the spectral properties of the noise[4;6]. Previous studies showed that the spatial correlation of the noise may impact diagnostic tasks, such as the detection of microcalcifications[7;3], highlighting the importance of incorporating this degradation into image processing.

Several studies have proposed model-based denoising, aiming to attenuate noise while preserving image quality in DM[8;9;10]. Although quantum and electronic noise components are frequently incorporated into these methods, the specific impact of spatial noise correlation on the performance of model-based algorithms in digital mammography has not yet been systematically investigated.

This study proposes and evaluates a model-based denoising framework for DM that explicitly accounts for spatially correlated noise. The main objective is to obtain denoised images that closely match the noise-free reference in both spatial and spectral domains while preserving microcalcification detectability. Two denoising strategies were investigated: one assuming spatially uncorrelated noise and another incorporating spatial noise correlation. For validation, synthetic images from virtual clinical trials (VCTs) were generated, and correlated noise was inserted into the noise-free images across multiple dose levels. The performance of the proposed approaches was assessed using the Mean Normalized Squared Error (MNSE), including its decomposition into bias and residual noise components, as well as spectral analysis through the Power Spectrum (PS). Additionally, a task-based evaluation was performed using a Channelized Hotelling Observer (CHO) to assess microcalcification detectability.

2 METHODS

2.1 Correlated noise model

In this work, noisy images were generated using a spatially correlated noise model. Considering Z as the correlated noisy image, the model can be defined as follows[4]:

$$Z(i) = \gamma Y(i) + \sigma(Y(i))\tilde{n}(i) + \tau, \quad \tilde{n}(i) = (n * \kappa_n)(i), \quad n(i) \sim \mathcal{N}(0,1), \quad \kappa_n = \frac{\kappa}{\|\kappa\|_2}, \tag{1}$$

where Y corresponds to the noise-free image, with i denoting the spatial pixel location; γ is the dose-scaling factor utilized to simulate different radiation dose levels. The dose-scaling factor ($0 < \gamma \leq 1$) enables the representation of noisy images under different dose conditions (e.g., 50%, 75%, and 100% AEC) within the same noise modeling framework; σ is the standard deviation function of the noise, and $\tilde{n}$ is a correlated Gaussian noise field with zero mean and unit variance. The term $\tilde{n}$ is obtained by convolving a Gaussian i.i.d. noise field n with the kernel κ_n, which introduces spatial correlation into the noise model. The kernel κ_n is normalized by $\|\kappa\|_2$ to ensure that $var\{\tilde{n}\} = 1$, thereby preserving the local variance after convolution. Finally, τ represents a constant offset added to avoid negative pixel values in the final image. The expectation and variance of Z are given by[4]:

$$E\{Z|Y\}(i) = \gamma Y(i) + \tau, \quad var\{Z|Y\}(i) = \xi_q(i)\left(\gamma Y(i)\right) + \xi_e^2, \tag{2}$$

where the expectation is related to the scaled noise-free image γY and the offset τ, while the variance is signal-dependent and determined by the quantum noise component ξ_q, which varies spatially with pixel position i and is associated with the system gain, and by the electronic noise component ξ_e, related to the detector readout process.

2.2 Simulation of noise images

Figure 1 shows the pipeline used to generate simulated correlated DM images Z from a synthetic noise-free image Y, as well as the two model-based denoising pipelines evaluated in this work. The simulation and denoising framework implemented in this study is publicly available at: GitHub LAVI-USP repository.

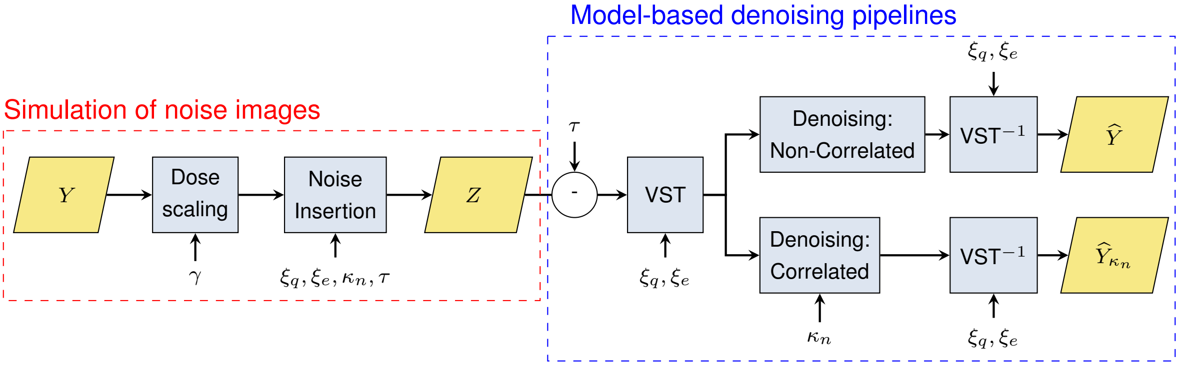

Figure 1: The main pipeline used in this work to simulate and denoise synthetic mammography images.

The simulation of Z initially requires the noise-free image Y. In this work, Y was obtained through the OpenVCT framework[11] and adjusted to simulate the exposure conditions of the GE Pristina system operating in DM mode. The reference image used to define the acquisition exposure was a clinical image from Barretos Cancer Hospital (Barretos, Brazil), acquired at 34 kVp and 50 mAs. These exposure parameters were considered as 100% AEC in this study.

After generating Y, three dose levels were simulated to evaluate the denoising approaches under different acquisition conditions: 50%, 75%, and 100% of AEC. These conditions were obtained by multiplying the noise-free image Y using the dose scaling factor γ (e.g., $\gamma = 0.5$ for 50% AEC).

Next, the noise insertion procedure was performed using the parameters ξ_q, ξ_e, κ_n, and τ. These parameters were estimated from uniform images acquired on the GE Pristina system in DM mode at different dose levels, following the protocol described in 4. All estimated parameters were then used to simulate and insert correlated noise into the noise-free image Y, generating the noisy image Z according to the model described in Eqs. (1–2), thereby mimicking GE Pristina acquisitions in DM mode. For each dose level, 20 independent noise realizations were generated.

2.3 Model-based denoising pipelines

Figure 1 also provides an overview of the denoising process utilized in this work. Two denoising strategies were evaluated, differing exclusively in how the noise was assumed to be modeled during the denoising step. All other processing stages were identical for both approaches.

To address the signal dependence of the noise model, a variance-stabilizing transform (VST) was applied to reduce its signal dependence. Specifically, the Generalized Anscombe Transform (GAT) was used[12]. After transformation, the noise can be approximated as Gaussian with unit variance and approximately signal-independent, enabling the application of Gaussian denoising algorithms.

In the denoising step, the two strategies differed in the model-based assumption. The first approach assumed spatially uncorrelated noise (above pipeline in Figure 1). The second approach explicitly incorporated spatial noise correlation by introducing the kernel κ_n as an additional parameter during filtering (bottom pipeline in Figure 1). In both cases, the BM3D was employed[13]. For the non-correlated assumption, BM3D was applied directly in the stabilized domain. For the correlated model-based approach, the BM3D algorithm was parameterized with the kernel (κ_n) to explicitly account for spatial noise correlation during the filtering process. After denoising, the inverse VST^{-1} was implemented using the inverse Generalized Anscombe Transform[12], restoring the images to the original range and resulting in the $\widehat{Y}$ (non-correlated noise assumption) and $\widehat{Y}_{\kappa_n}$ (spatial correlation noise assumption).

2.4 Validation of denoised images

The $\widehat{Y}$ and $\widehat{Y}_{\kappa_n}$ were compared with the noise-free reference image (Y) across all simulated dose levels. The denoising performance was evaluated using MNSE, which enables the assessment of denoising performance in terms of signal loss and noise removal[9]. In particular, MNSE was decomposed into bias (signal loss), quantified by $\mathcal{B}^2$, and residual noise, represented by $\mathcal{R}_n$, allowing a more detailed analysis of the trade-off between noise removal and signal preservation. All these metrics were computed relative to the noise-free reference image (Y), and their values are reported as percentages with respect to this ground truth.

Additionally, detection performance was evaluated using a Channelized Hotelling Observer (CHO), with results reported as the detectability index (d'), computed from the separation between signal-present and signal-absent responses of the observer[14;15]. The CHO was applied to all image conditions (Z, Y, $\widehat{Y}$, and $\widehat{Y}_{\kappa_n}$) to assess the detectability of microcalcifications. A total of 100 ROIs (170 $\times$ 170 px) with and without microcalcifications were used for training, and an independent set of 100 ROIs was used for testing. The channel width parameter for CHO was set to 2.5.

It is important to note that the datasets used for MNSE evaluation and model observer analysis differ. While the MNSE calculations were performed using background images generated from VCT, the CHO analysis required signal-present and signal-absent cases. For this purpose, a microcalcification cluster composed of five calcifications (310μm size $\approx$ 3 px) was inserted into the noise-free VCT images (Y), using the framework described in 16, with randomized spatial locations to account for background variability. The microcalcification contrast utilized during the insertion was 8%.

To further assess the impact in the spectral domain, the denoised images were analyzed using power spectrum (PS) analysis[7;4], comparing the noise-free reference (Y), the non-correlated noise ($\widehat{Y}$), and the correlation-based ($\widehat{Y}_{\kappa_n}$) assumptions.

3 RESULTS AND DISCUSSION

The noise parameters estimated in this work are: $\xi_e = 2.8$, and $\tau = 1.9$. Figure 2 shows the quantum gain map ξ_q and the normalized correlation kernel κ_n.

Figure 3 shows the cropped regions from Y, Z, and the denoised results obtained using the non-correlated noise ($\widehat{Y}$) and the correlation-based ($\widehat{Y}_{\kappa_n}$) assumptions across all radiation dose levels. As expected, noise degradation increases with decreasing radiation dose. The denoising approach assuming uncorrelated noise effectively suppresses noise. However, it introduces smoothing, leading to a loss of fine structural details. In contrast, the correlation-based approach better preserves image details while still achieving substantial noise reduction.

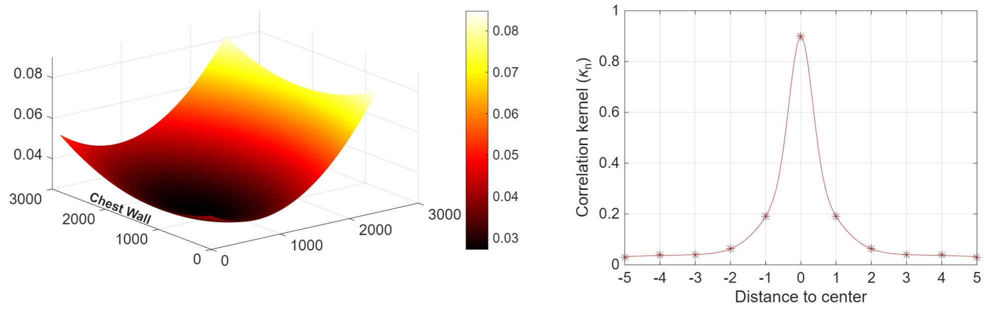

Figure 2: Estimated parameters: quantum gain map ξ_q (left), and Normalized convolution kernel κ_n (right).

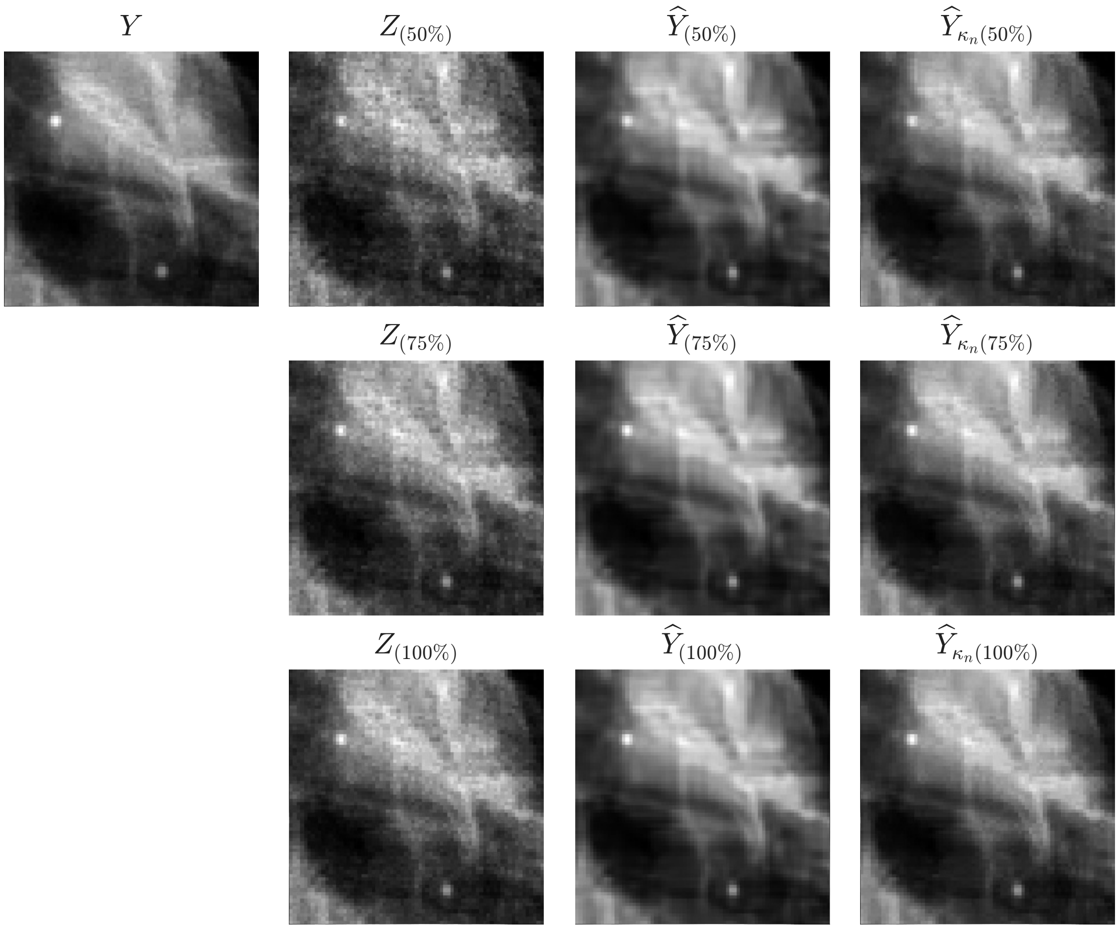

Figure 3: Cropped regions of simulated images across different radiation doses and denoised conditions.

Table 1 summarizes the results for all evaluated methods. Overall, the correlation-based approach consistently achieved superior performance compared to the non-correlated noise assumption. In terms of MNSE, incorporating spatial noise correlation led to lower error values at all dose levels, with the most significant improvement observed at 50% AEC. This gain is primarily driven by a substantial reduction in the bias component ($\mathcal{B}^2$), which indicates significantly improved signal preservation for the correlation-based approach. Notably, this reduction in bias was sufficient to compensate for the slight increase in residual noise ($\mathcal{R}_n$) observed at higher dose levels (75% and 100% AEC), still resulting in a lower total MNSE compared to the non-correlated method. Regarding the detectability index (d'), the correlation-based approach also outperformed the non-correlated assumption across all doses.

Table 1: Results of MNSE, $\mathcal{R}_n$, $\mathcal{B}^2$, and detectability index (d') for the noise (Z) and denoised images ($\widehat{Y}$,$\widehat{Y}_{\kappa_n}$). Confidence intervals are shown at a significance level of $p = 0.05$. The noise-free reference image (Y) achieved $d' = 12.21$, representing the upper bound of detectability.

Radiation dose	Metrics (%)	Noise image (Z)	Non-correlated noise assumption ($\widehat{Y}$)	Correlated noise assumption ($\widehat{Y}_{\kappa_n}$)
50%	MNSE	9.67 [9.66, 9.69]	4.76 [4.75, 4.78]	**4.50 [4.48, 4.51]**
	$\mathcal{R}_n$	9.67 [9.67, 9.68]	3.80 [3.80, 3.80]	3.77 [3.77, 3.78]
	$\mathcal{B}^2$	0.0 [0.0, 0.0]	0.97 [0.96, 0.97]	0.72 [0.72, 0.73]
	d'	8.38	9.27	**9.41**
75%	MNSE	5.69 [5.68, 5.70]	3.28 [3.28, 3.29]	**3.08 [3.07, 3.08]**
	$\mathcal{R}_n$	5.69 [5.68, 5.69]	2.49 [2.49, 2.49]	2.56 [2.55, 2.56]
	$\mathcal{B}^2$	0.0 [0.0, 0.0]	0.79 [0.79, 0.80]	0.52 [0.52, 0.52]
	d'	9.32	10.04	**10.37**
100%	MNSE	3.98 [3.97, 3.99]	2.56 [2.56, 2.57]	**2.37 [2.37, 2.38]**
	$\mathcal{R}_n$	3.98 [3.98, 3.98]	1.89 [1.89, 1.89]	1.96 [1.96, 1.97]
	$\mathcal{B}^2$	0.0 [0.0, 0.0]	0.68 [0.67, 0.68]	0.41 [0.41, 0.41]
	d'	9.90	10.41	**10.78**

The detectability index (d') obtained using the CHO consistently increased when spatial correlation was incorporated into the denoising process. Across all dose levels, the correlation-based approach achieved the highest d' values, indicating improved detectability of microcalcifications. This trend is consistent with the improvements observed in MNSE and indicates that better noise modeling leads not only to improved image quality but also to enhanced detection performance.

Figure 4 presents the PS analysis for all simulated images. As expected, the noisy images (Z) exhibit higher PS across the frequency range compared to the other image conditions at all evaluated radiation dose levels.

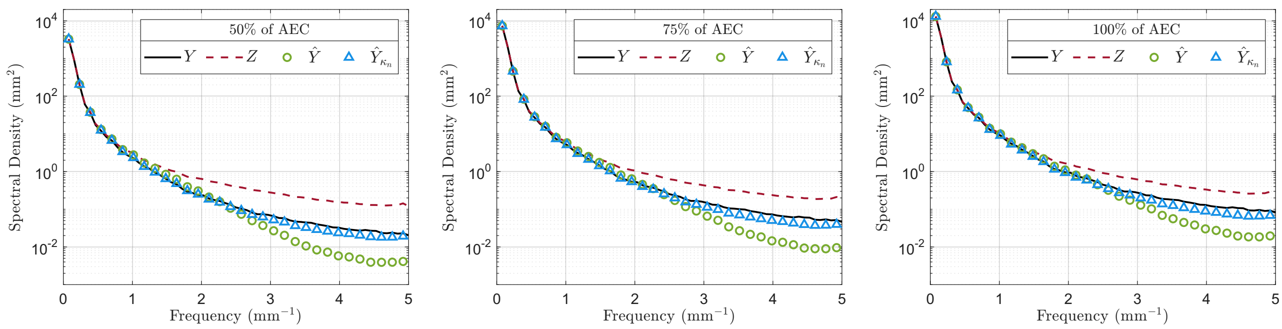

Figure 4: PS of Y, Z, $\widehat{Y}$, $\widehat{Y}_{\kappa_n}$ for the 50%, 75%, and 100% of AEC, respectively.

After denoising, both approaches reduce the PS magnitude. However, the correlation-based approach yields a power spectrum that more closely matches the reference image (Y) across the entire frequency range, indicating better preservation of the signal. In contrast, the non-correlated noise approach tends to have more impact at higher frequencies, suggesting over-smoothing and loss of fine details. These findings are consistent with the visual results shown in Figure 3.

4 CONCLUSIONS

This study showed that incorporating spatial noise correlation into a model-based denoising pipeline improves restoration performance in DM images. The proposed approach achieved lower MNSE values, improved residual noise suppression with less signal loss, and better spectral match with the noise-free reference image, particularly at higher spatial frequencies. From a task-based perspective, the incorporation of spatial correlation also led to higher detectability index (d') values, indicating improved microcalcification detection performance. These findings suggest that accurate spectral noise modeling may be essential for preserving diagnostically relevant high-frequency structures in low-dose mammography. Future work will extend this analysis to more complex task-based detection scenarios, including studies involving human observers and different microcalcification sizes.

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